Tien-Chen Lin, Ph.D.

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Research profile

I investigate how cytoskeletal dynamics establish neuronal polarity and how the resulting compartment identities become durable. My research spans molecular mechanisms of microtubule nucleation and the whole-cell dynamics that coordinate neurite growth and axon selection. My first-author Nature study (2026) identified a perisomatically organized ARP2/3-myosin-II oscillatory program that coordinates recurrent neurite growth and retraction during axon selection. Previously, I independently initiated the MOZART1 project and reconstituted an active microtubule-nucleation template (Journal of Cell Biology, 2016), building on my doctoral work defining phosphorylation-dependent regulation of mitotic spindle assembly (eLife, 2014). My independent research program will determine when axonal identity becomes durable, how competing neurites are restricted from sustaining additional axons, and how developing tissue influences persistence and selective reorganization. I combine quantitative live imaging, spatial perturbation and mechanistic biochemistry to connect whole-cell behavior with molecular cytoskeletal organization.

Education

Ph.D., Natural Sciences (magna cum laude, Dr. rer. nat.) 2009-2014
University of Heidelberg, Germany

M.S., Biochemistry and Molecular Biology (5-year B.S./M.S. fast-track program) 2005-2006
National Yang-Ming University (now National Yang Ming Chiao Tung University), Taiwan

B.S., Life Sciences 2002-2005
National Yang-Ming University, Taiwan

Research experience

Postdoctoral fellow 2017-present
German Center for Neurodegenerative Diseases (DZNE)
Supervisor: Prof. Dr. Frank Bradke

- Conceptually reframed and led the study identifying a perisomatically organized ARP2/3-myosin-II oscillatory program that coordinates neurite growth during neuronal polarization (first author, Nature, 2026).
- Built quantitative and spatial-perturbation capability for the project, including R/ImageJ analysis pipelines and, with Andreas Husch, re-establishment of local single-cell perfusion for acute spatial manipulation.
- Initiated quantitative modeling of neuronal polarization to compare alternative dynamical explanations and generate experimentally testable predictions.
- Built multidisciplinary collaborations spanning electron tomography, quantitative image analysis and acute molecular perturbation to resolve different levels of the polarity mechanism.
- Contributed quantitative transcriptomic analysis to collaborative projects in CNS regeneration and injury, including work published in Neuron and Brain Communications in 2022, and mapped injury responses across dorsal-root-ganglion neuron subtypes.

Research experience continued

Postdoctoral fellow 2014-2017

Zentrum für Molekulare Biologie Heidelberg (ZMBH), University of Heidelberg

Supervisor: Prof. Dr. Elmar Schiebel

- Developed the cross-species analysis of γ-tubulin-complex targeting (corresponding-author Trends in Cell Biology review, 2015) that led to the MZT1/MOZART1 project in Candida albicans.
- Secured an EMBO Short-Term Fellowship to collaborate with Robert Arkowitz and established recombinant purification, microtubule-nucleation and quantitative binding assays.
- Showed that MOZART1 and γ-tubulin-complex receptors assemble γ-TuSC into an active microtubule-nucleation template (first author, J. Cell Biol., 2016); the platform remained in use after departure.

Doctoral researcher 2009-2014

Zentrum für Molekulare Biologie Heidelberg (ZMBH), University of Heidelberg

Supervisor: Prof. Dr. Elmar Schiebel

- Defined phosphoregulation of γ-TuSC and its receptor controlling mitotic microtubule nucleation (first-author eLife, 2014; PLoS ONE, 2011).
- Established the physical interaction between γ-TuSC and its microtubule-organizing-center receptor.
- Identified conserved relationships among divergent γ-tubulin-complex targeting factors, providing the basis for later comparative work.

Research assistant 2007-2009

Genomics Research Center, Academia Sinica, Taipei, Taiwan

Supervisor: Prof. Dr. Ming-Daw Tsai

- Structural characterization of HICE1, HEC1 and HACE1.
- Biochemical characterization of the GRP78-NPGPx interaction.

Additional experience

Compulsory military service, Taiwan 2006-2007

Selected first-author publications

2026 Lin, T.-C. et al. Nature 657, 213-225 - cytoskeletal oscillations during axon selection.

2016 Lin, T.-C. et al. J. Cell Biol. 215, 823-840 - reconstitution of an active microtubule-nucleation template.

2014 Lin, T.-C. et al. eLife 3, e02208 - phosphorylation-dependent regulation of γ-TuSC nucleation.

Full publication list: tien-chen-lin.github.io/publications

Supervision and mentoring

- Supervision of the research rotation of M.Sc. student Josephine Hahn, DZNE Bonn 2025. Designed and supervised a one-month live-imaging project on ARP2/3/myosin-II control of microtubule retrograde flow, including pharmacological perturbation, Fiji kymographs and R analysis.
- Supervision of research project of M.Sc. student Annalena Meyer, University of Heidelberg 2017. Project: Investigation of the GRIP1 domain of γ-tubulin complex subunit GCP3.
- Supervision of research project of M.Sc. student Yvonne Schlosser, University of Heidelberg 2014. Project: Investigation of the N-terminal domain of γ-tubulin complex subunit GCP3.

Teaching service and professional development

Initiator and coordinator, FRM 2027 symposium proposal on neuronal polarity, Istanbul 2026-2027

Associate Faculty Member, H1 Connect (formerly F1000 Faculty Opinions) 2019-present

University of Bonn

Coordinator of M.Sc. program "Molecular Cell Biology" EM5 practical module 2023-2025

M.Sc. "Molecular Cell Biology" EM5 module, lecturer of Data Visualization with R (4 hours) 2023

M.Sc. "Molecular Cell Biology" EM5 module, lecturer of Light Microscopy Basics (2 hours) 2022

M.Sc. "Neuroscience", teacher of Live-cell Imaging session (2 days) 2022

M.Sc. "Molecular Cell Biology" EM5 module, lecturer of Light Microscopy Basics (2 hours) 2020

University of Heidelberg

M.Sc. practical course "Mechanism and Regulation of Eukaryotic Cell Division" (1 week) 2013

M.Sc. lecture "Cell Cycle Checkpoint Regulation" (2 hours) 2012

Leadership and professional training

Laboratory Leadership for Group Leaders, EMBO 2020

Project Management for Scientists, EMBO 2021

Negotiation for Scientists, EMBO 2020

Advanced Techniques in Molecular Neuroscience, Cold Spring Harbor Laboratory 2020

Fellowships

EMBO Short-term Fellowship for C. albicans γ-tubulin small complex and MOZART1 project 2016

LGFG Fellowship (Landesgraduiertenförderung-Programme) "Regulation of Cell Division" 2012

Talks

Invited seminar (scheduled 12 Nov 2026), Tsinghua University, Beijing, China 2026

Invited seminar (15 Oct 2026), FAU Erlangen-Nürnberg, Erlangen 2026

Microtubules in Neurons, Chiemsee 2023

The Cytoskeleton of Neurons and Glia webinar series, virtual 2023

EMBO Workshop Neural development and neurodegeneration, Taipei 2022

FENS Symposium Barriers and hopes for functional restoration after traumatic CNS injury, Paris 2022

CSHL meeting Molecular Mechanisms of Neuronal Connectivity, New York 2022

European Cytoskeleton Forum, Hannover 2022

Seminar, International Center for Wound Repair and Regeneration, College of Medicine, National Cheng Kung University, Tainan 2022

Seminar, Department of Biological Science and Technology, National Yang Ming Chiao Tung University, Hsinchu 2022

DKFZ-ZMBH Alliance Young Investigators Seminar, Heidelberg 2016